Practical Week 2 -Task 1&2

Step 1: Open Ubuntu

Open Ubuntu from the Windows Start Menu.

You should see:

jhanasri@LAPTOP-1RN1M8E3:~$

Step 2: Go to Your Project Folder

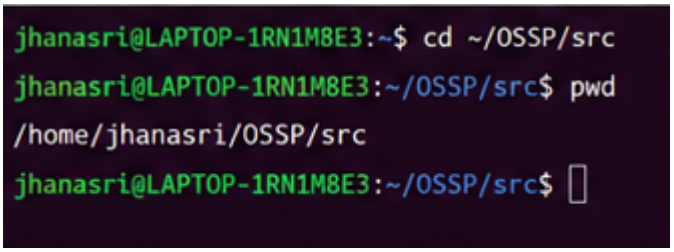
cd ~/OSSP/src

Check the current directory:

pwd

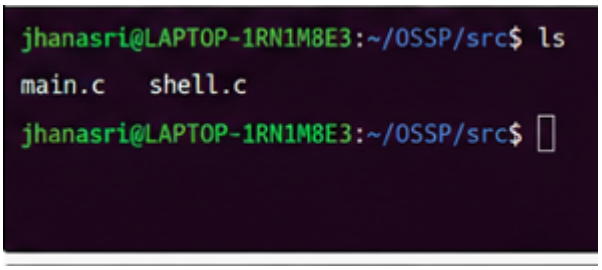
Expected output:

/home/jhanasri/OSSP/src



Step 3: View Existing Files

Ls



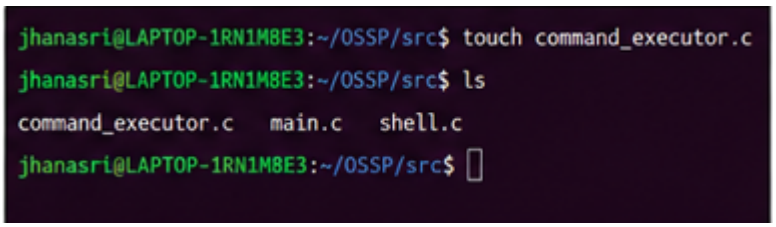
Step 4: Create a New C File

touch command_executor.c

Verify:

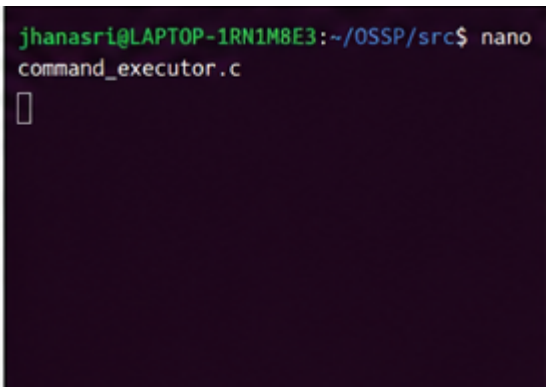
ls

You should see `command_executor.c`.



Step 5: Open the File

nano command_executor.c

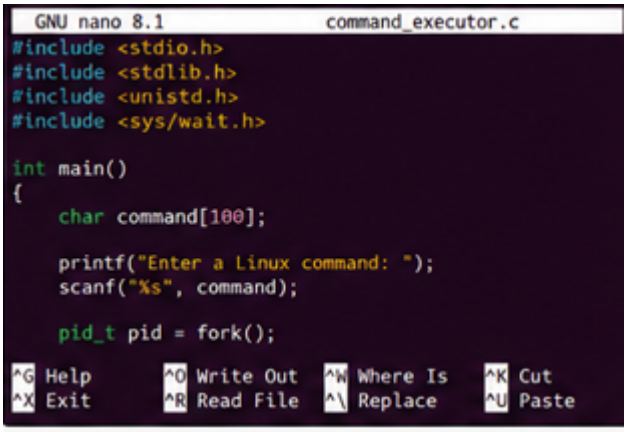


Step 6: Type/Paste the Program

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0)
    {
        printf("Fork Failed!\n");
        return 1;
    }
    else if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
```

```
        printf("Parent PID : %d\n", getppid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);

        printf("Command not found\n");
    }
    else
    {
        wait(NULL);

        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        printf("Child process completed.\n");
    }

    return 0;
}
```

This code performs the operations required in Task 1 of your document.



Step 7: Save the File

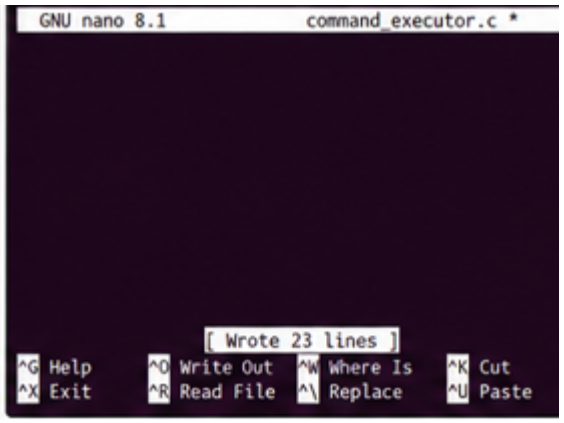
Press:

Ctrl + O

Press Enter

Then press:

Ctrl + X

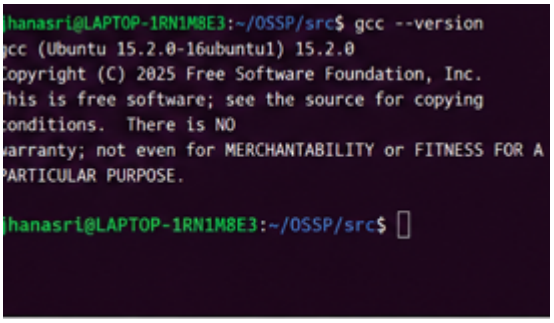


Step 8: Check GCC

gcc --version

If GCC is not installed:

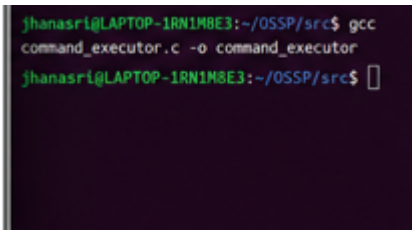
sudo apt update
sudo apt install build-essential -y



Step 9: Compile the Program

gcc command_executor.c -o command_executor

If there are no errors, the compilation is successful.

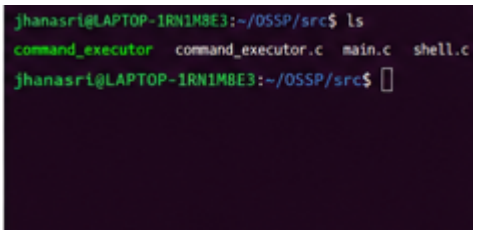


Step 10: Verify the Executable

ls

You should see:

command_executor



Step 11: Run the Program

./command_executor

Step 12: Test with `ls`

When prompted:

Enter a Linux command:

Type:

ls

Example output:

Enter a Linux command: ls

Child Process
Child PID : 1234
Parent PID : 1233
Executing command: ls

command_executor
command_executor.c
main.c
shell.c

Parent Process
Parent PID : 1233
Child PID : 1234
Child process completed.

Step 13: Test with `pwd`

Run again:

./command_executor

Input:

Pwd

-

```
jhanasri@LAPTOP-1RN1MBE3:~/OSSP/src$ ./command_executor
Enter a Linux command: pwd

Child Process
Child PID  : 12367
Parent PID  : 12366
Executing command: pwd

/home/jhanasri/OSSP/src

Parent Process
Parent PID  : 12366
Child PID  : 12367
Child process completed.

jhanasri@LAPTOP-1RN1MBE3:~/OSSP/src$
```

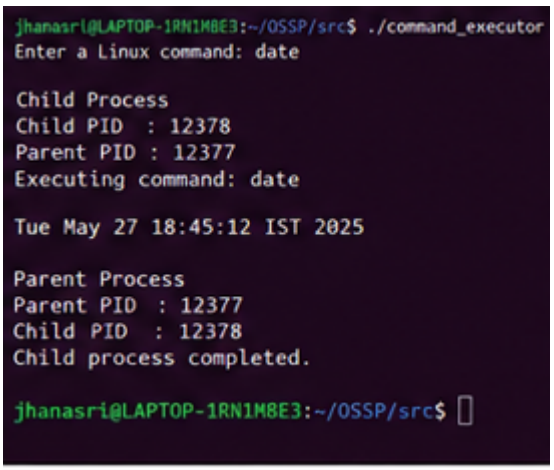
Step 14: Test with `date`

Run again:

./command_executor

Input:

Date



Step 15: Test with `whoami`

Run again:

./command_executor

Input:

whoami

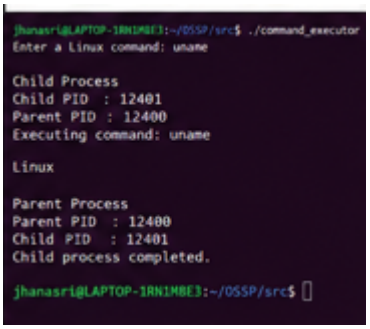
Step 16: Test with `uname`

Run again:

./command_executor

Input:

Uname



Step 17: Verify the Requirements

Check that your program

Practical Week 2 - Task 2

Step 1: Open Ubuntu

Open Ubuntu from the Start Menu.

You should see:

jhanasri@LAPTOP-1RN1M8E3:~$

Step 2: Go to Home Directory

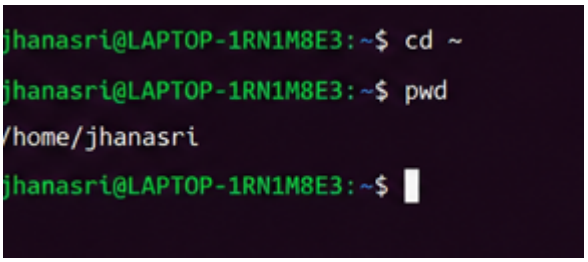
cd ~

Check:

pwd

Expected:

/home/jhanasri



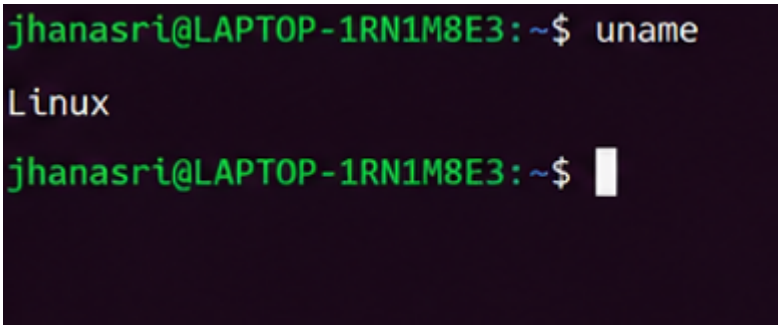
Step 3: Display Linux Kernel Information

Run:

uname

Example Output:

Linux



Step 4: Display Complete System Information

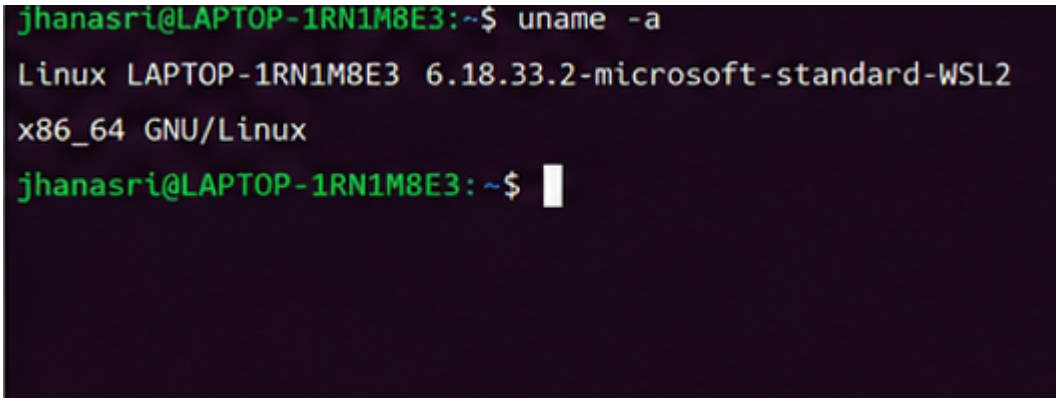
Run:

uname -a

Example Output:

Linux LAPTOP-1RN1M8E3 6.18.33.2-microsoft-standard-WSL2 x86_64 GNU/Linux

Take a screenshot.



Step 5: Display CPU Information

Run:

lscpu

Example Output:

Architecture: x86_64
CPU(s): 8
Vendor ID: GenuineIntel
Model name: Intel(R) Core(TM)

Take a screenshot.

```
Architecture:          x86_64
CPU(s):                8
On-line CPU(s) list:   0-7
Vendor ID:             GenuineIntel
Model name:            Intel(R) Core(TM) i5-1135G7
CPU MHz:               2400.000
Cache size:            8192 KB
Flags:                 fpu vme de pse tsc msr pae mce cx8
                       apic sep mtrr pge mca cmov pat pse36
                       clflush dts acpi mmx fxsr sse sse2
jhanasri@LAPTOP-1RN1M8E3:~$
```

Step 6: Display Running Processes

Run:

ps

Example Output:

PID TTY TIME CMD
1023 pts/0 00:00:00 bash
1050 pts/0 00:00:00 ps

Take a screenshot.

```
jhanasri@LAPTOP-1RN1M8E3:~$ ps
  PID TTY          TIME CMD
 2480 pts/0    00:00:00 bash
 2532 pts/0    00:00:00 ps
jhanasri@LAPTOP-1RN1M8E3:~$
```

Step 7: Display Detailed Running Processes

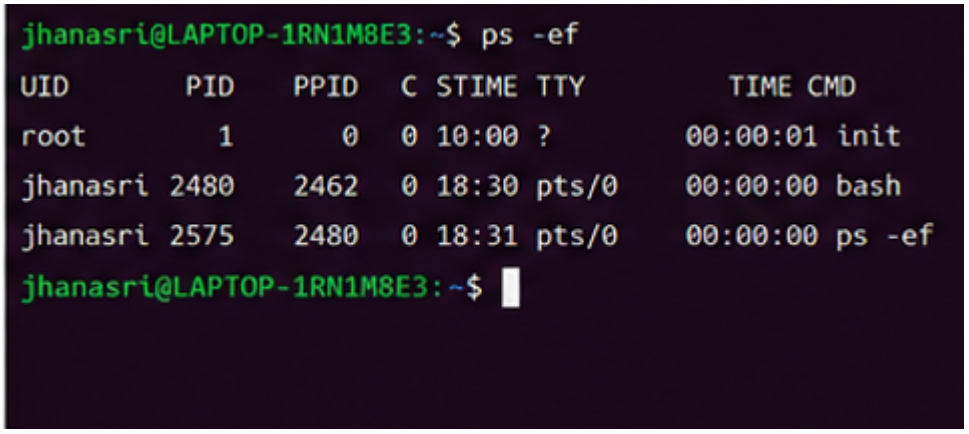
Run:

ps -ef

Example Output:

UID PID PPID C STIME TTY TIME CMD
root 1 0 0 10:00 ? 00:00:00 init
user 1050 1023 0 10:10 pts/0 00:00:00 bash

Take a screenshot.



Step 8: Display Live System Processes

Run:

top

Example Screen:

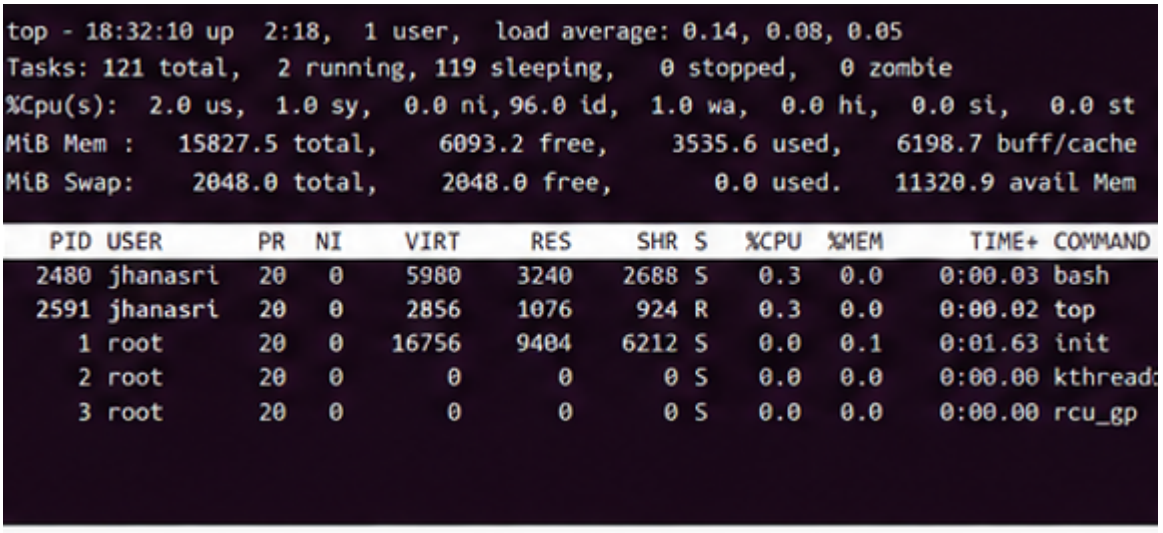
top - 18:45:10

Tasks: 120 total

CPU: 5%

Memory: 22%

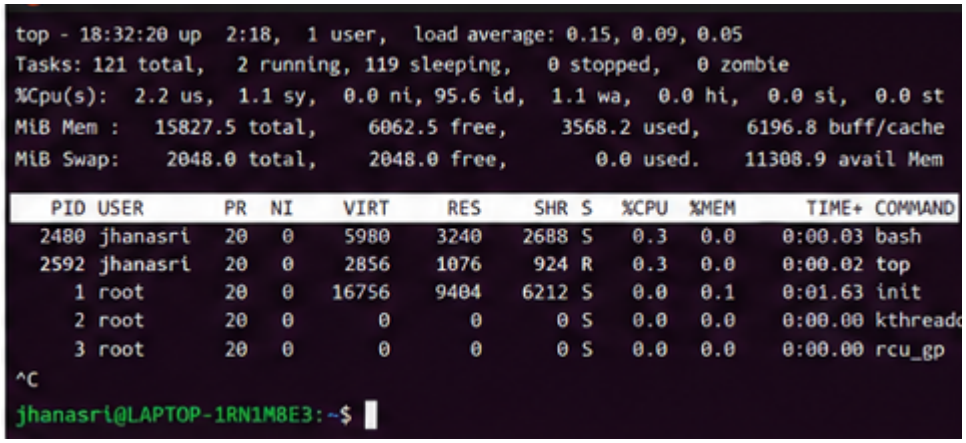
Take a screenshot.



Step 9: Exit Top

Press

Q



Step 10: CPU Abstraction

Write in your record:

> The operating system manages CPU time by scheduling different processes so that each program gets a fair share of processor time.

Step 11: Memory Abstraction

Write:

> The operating system allocates memory to each process and protects one process from accessing another process's memory.

Step 12: Storage Abstraction

Write:

> The operating system manages files and directories and provides a file system for storing and retrieving data.

Step 13: I/O Device Abstraction

Write:

> The operating system communicates with hardware devices such as the keyboard, mouse, monitor, printer, and storage devices through device drivers.

Step 14: Observation

Write:

> The Linux commands successfully displayed kernel information, CPU details, running processes, and live system resource usage. The operating system abstracts hardware resources and provides services to user applications.

Step 15: Result

Write:

> The Linux terminal commands `uname`, `lscpu`, `ps`, and `top` were executed successfully. The relationship between hardware resources and operating system services was studied, and the abstraction of CPU, memory, storage, and I/O devices was understood.